Chapter 12, Problem 32P

Problem

Design a problem to help other students better understand power in a balanced three-phase system.

Step-by-step solution

Step 1 of 2

Design a problem to help other students better understand power in a balanced three-phase system.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

A balanced wye load is connected to a 60-Hz three-phase source with $V_{ab} = 240\angle 0^\circ$V. The load has lagging pf =0.5 and each phase draws 5 kW. (a) Determine the load impedance Z_Y. (b) Find I_a, I_b, and I_c.

Solution

(a) $|V_{ab}| = \sqrt{3}V_p = 240 \longrightarrow V_p = \frac{240}{\sqrt{3}} = 138.56$

$$V_{an} = V_p < -30^\circ$$

$$pf = 0.5 = \cos\theta \longrightarrow \theta = 60^\circ$$

$$P = S\cos\theta \longrightarrow S = \frac{P}{\cos\theta} = \frac{5}{0.5} = 10\text{ kVA}$$

$$Q = S\sin\theta = 10\sin 60 = 8.66$$

$$S_p = 5 + j8.66\text{ kVA}$$

But

$$S_p = \frac{V_p^2}{Z_p^*} \longrightarrow Z_p^* = \frac{V_p^2}{S_p} = \frac{138.56^2}{(5+j8.66)x10^3} = 0.96 - j1.663$$

Zp = 0.96 + j1.663Ω

Step 2 of 2

(b) $I_a = \frac{V_{an}}{Z_Y} = \frac{138.56 < -30^\circ}{0.96 + j1.6627} = \underline{72.17 < -90^\circ\text{ A}}$

$$I_b = I_a < -120^\circ = \underline{72.17 < -210^\circ\text{ A}}$$

$$I_c = I_a < +120^\circ = \underline{72.17 < 30^\circ\text{ A}}$$